Theory Notes (From Slides)

1. Greedy Technique

Introduction

The greedy method is a problem-solving strategy where decisions are made by selecting the best option available at the moment without considering future consequences. It builds a solution step by step by always choosing the immediate best option. Greedy algorithms are mainly used for optimization problems.

Properties of Greedy Algorithms

Greedy Choice Property

The optimal solution can be constructed by making the best local choice at each step.

Optimal Substructure

The optimal solution to a problem contains the optimal solutions to its subproblems.

Working Procedure of Greedy Algorithm

1. Start with the initial state.
2. Consider all available choices.
3. Select the best choice at that moment.
4. Move to the next state.
5. Repeat until the goal is achieved.

Limitation of Greedy Algorithm

Greedy algorithms do not always produce the optimal solution. In problems such as Coin Change and 0/1 Knapsack, Dynamic Programming gives better results.

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2. Fractional Knapsack Problem

Definition

Given arrays of profits and weights and a knapsack capacity W, the goal is to maximize total profit without exceeding capacity. Items can be taken fractionally if needed.

Greedy Approaches

- Selecting smaller weights first
- Selecting larger profits first
- Selecting by profit/weight ratio

The profit/weight ratio method gives the best result.

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3. Minimum Spanning Tree (MST)

Spanning Tree

A spanning tree of a graph is a subgraph that:

- Contains all vertices
- Is itself a tree

A graph may contain many spanning trees.

Minimum Spanning Tree

The Minimum Spanning Tree is the spanning tree with the minimum total cost.

Algorithms for MST

- Kruskal’s Algorithm
- Prim’s Algorithm

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4. Kruskal’s Algorithm

Definition

Kruskal’s algorithm creates a forest of trees. Initially, every node is considered as a separate tree. The algorithm repeatedly adds the cheapest edge that does not create a cycle.

Steps of Kruskal’s Algorithm

1. Construct a forest with each node in a separate tree.
2. Place all edges in a priority queue.
3. Extract the minimum cost edge.
4. If it forms a cycle, reject it.
5. Otherwise, add it to the forest.
6. Continue until $n-1$ edges are added.

Important Point

Edges creating cycles are not added to the spanning tree.

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5. Single Source Shortest Path (SSSP)

Definition

The shortest path problem finds the minimum cost path between nodes in a graph.

Path

A path is a sequence of vertices such that every consecutive pair of vertices has an edge between them.

Cost of a Path

In weighted graphs, the cost of a path is the sum of all edge weights along that path.

Variations of Shortest Path

- Single-pair shortest path
- Single-source shortest path

- Single-destination shortest path
- All-pair shortest path

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6. Dijkstra’s Algorithm

Definition

Dijkstra’s algorithm is used for solving single-source shortest path problems in weighted graphs. It uses a greedy approach.

Characteristics

- Used in maps and navigation systems
- Works for directed and undirected graphs
- Does not work with negative edge weights

Working Procedure

1. Mark all nodes as unvisited.
2. Set source distance to 0 and others to infinity.
3. Select the unvisited node with minimum distance.
4. Update distances of neighboring nodes.
5. Mark the current node as visited.
6. Repeat until all nodes are visited.

Relaxation Condition

$d(u)+c(u,v)<d(v)$

Then update:
$d(v)=d(u)+c(u,v)$

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7. Bellman-Ford Algorithm

Definition

Bellman-Ford algorithm is used to find shortest paths in graphs containing negative edge weights.

Main Idea

The algorithm repeatedly relaxes all edges.

Relaxation Formula

$D[U]+D[U,V]<D[V]$

If true:
$D[V]=D[U]+D[U,V]$

Steps of Bellman-Ford Algorithm

1. Initialize all distances as infinity.
2. Set source distance to 0.
3. Relax all edges $V-1$ times.
4. Check all edges once more.
5. If any distance changes in the V-th iteration, a negative cycle exists.

Negative Weight Cycle

A change in the V-th iteration indicates the presence of a negative weight cycle.

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8. Topological Sort

Definition

Topological sorting is a linear ordering of vertices in a Directed Acyclic Graph (DAG) such that if there is an edge $u \rightarrow v$, then u appears before v.

Conditions

Topological sorting is only possible for Directed Acyclic Graphs (DAGs).

Why Not Possible for Undirected or Cyclic Graphs

Undirected Graph

Every edge can be treated in both directions, so ordering becomes impossible.

Cyclic Graph

In a cycle like u → v → w → u, dependencies become impossible to satisfy.

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9. Kahn’s Algorithm

Definition

Kahn’s Algorithm is a BFS-based method for topological sorting.

Steps

1. Calculate indegree of every vertex.
2. Insert all vertices with indegree 0 into a queue.
3. Remove a vertex from the queue and add it to the result.
4. Reduce indegree of adjacent vertices.
5. If indegree becomes 0, insert into queue.
6. Continue until queue becomes empty.

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10. DFS Based Topological Sort

Main Idea

DFS traversal is performed and vertices are pushed into a stack during backtracking.

Steps

1. Start DFS from an unvisited node.
2. Visit all neighbors recursively.
3. Push the node into stack after visiting all neighbors.
4. Pop elements from stack to get topological order.

11. Backtracking

Definition

Backtracking is an algorithmic technique where solutions are built incrementally and abandoned when they lead to invalid states or dead ends.

Working Procedure

- Choose
- Explore
- Check validity
- Backtrack
- Repeat

Applications

- Sudoku solving
- N Queens Problem
- Graph Coloring
- Robotic Path Planning

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12. Graph Coloring Problem

Definition

Graph coloring is assigning colors to graph vertices such that no two adjacent vertices have the same color.

m-Coloring

If at most m colors are used, it is called m-coloring.

Chromatic Number

The minimum number of colors needed to color a graph is called its chromatic number.

Applications

- Map coloring
- Timetable design

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13. N Queens Problem

Definition

The N-Queens problem places N queens on an N×N chessboard so that no two queens attack each other.

Conditions

Queens must not share:

- Same row
- Same column
- Same diagonal

Solution Idea

Queens are placed row by row. If a position becomes unsafe, the algorithm backtracks and tries another position.

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14. Branch and Bound

Definition

Branch and Bound is an optimization technique used to find optimal solutions for combinatorial and mathematical optimization problems.

Main Concepts

Branching

Breaking the main problem into smaller subproblems.

Bounding

Calculating upper or lower bounds for subproblems.

Pruning

Discarding branches that cannot produce better solutions.

Applications

- Discrete optimization
- Integer programming
- Network flow problems

FIFO Branch and Bound

Uses queue and performs Breadth First Search (BFS).

0/1 Knapsack Problem

Each item can either be taken completely or not taken at all.

Why Dynamic Programming is Better

For classic 0/1 knapsack:
O(n\times W)

Dynamic Programming guarantees an optimal solution.

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15. Approximation Algorithm

Definition

Approximation algorithms are algorithms designed to find near-optimal solutions for NP-hard problems.

Why Needed

Exact solutions for NP-hard problems are computationally infeasible in polynomial time.

Approximate Solution

A feasible solution close to the optimal solution is called an approximate solution.

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16. Traveling Salesman Problem (TSP)

Definition

The Traveling Salesman Problem requires visiting all cities exactly once and returning to the starting city with minimum total cost.

Characteristics

- NP-hard problem
- No polynomial-time exact solution exists

Triangle Inequality

$d(A,C)\le d(A,B)+d(B,C)$

Direct travel from A to C should not cost more than traveling through B.

TSP Approximation Steps

1. Choose a starting vertex.
2. Construct a Minimum Spanning Tree.
3. Perform preorder traversal.
4. Use traversal order as Hamiltonian path.

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17. Vertex Cover Problem

Definition

A vertex cover is a subset of vertices such that every edge in the graph has at least one endpoint in the subset.

Goal

Find the minimum size vertex cover.

Approximation Algorithm Steps

1. Select any edge randomly.
2. Add both endpoints to the output set.
3. Remove all incident edges.
4. Repeat until all edges are covered.

Applications

Used in traffic camera placement systems.

Short Questions and Answers (According to Slides)

Greedy Technique

Q1. What is a greedy algorithm?

A greedy algorithm is a problem-solving technique that chooses the best available option at each step without considering future consequences.

Q2. In which type of problems are greedy algorithms mainly used?

Greedy algorithms are mainly used in optimization problems.

Q3. What is the Greedy Choice Property?

It means the optimal solution can be achieved by making the locally best choice at every step.

Q4. What is Optimal Substructure?

Optimal substructure means the optimal solution contains optimal solutions of its subproblems.

Q5. Why do greedy algorithms fail in some cases?

Because choosing the locally optimal solution may not always produce the globally optimal solution.

Q6. Give examples where greedy algorithms fail.

- Coin Change Problem
- 0/1 Knapsack Problem

Q7. What are the applications of greedy algorithms?

- Fractional Knapsack
- Job Scheduling

Fractional Knapsack Problem

Q8. What is the Fractional Knapsack Problem?

It is a problem where items can be taken fractionally to maximize profit without exceeding capacity.

Q9. Which greedy strategy gives the best result in Fractional Knapsack?

Selecting items based on the profit/weight ratio gives the best result.

Q10. Can items be divided in Fractional Knapsack?

Yes, items can be taken partially or fractionally.

Minimum Spanning Tree (MST)

Q11. What is a spanning tree?

A spanning tree is a subgraph that contains all vertices of a graph and forms a tree.

Q12. What is a Minimum Spanning Tree?

A Minimum Spanning Tree is a spanning tree with the minimum total edge cost.

Q13. Name two MST algorithms.

- Kruskal's Algorithm
- Prim's Algorithm

Q14. How many edges does an MST contain?

An MST contains $n-1$ edges where n is the number of vertices.

Kruskal’s Algorithm

Q15. What is Kruskal’s Algorithm?

Kruskal’s Algorithm is a greedy algorithm used to find the Minimum Spanning Tree by selecting minimum cost edges without forming cycles.

Q16. What data structure is commonly used in Kruskal’s Algorithm?

Priority Queue is commonly used.

Q17. What happens if an edge forms a cycle in Kruskal’s Algorithm?

The edge is rejected.

Q18. Why is Kruskal’s Algorithm called greedy?

Because it always selects the minimum cost edge available.

Single Source Shortest Path (SSSP)

Q19. What is the shortest path problem?

It is the problem of finding the minimum cost path between nodes in a graph.

Q20. What is the cost of a path in a weighted graph?

The cost is the sum of all edge weights along the path.

Q21. Name the variations of shortest path problems.

- Single-pair shortest path
- Single-source shortest path
- Single-destination shortest path
- All-pair shortest path

Dijkstra’s Algorithm

Q22. What is Dijkstra’s Algorithm?

It is a greedy algorithm used to find the shortest path from a source node to all other nodes.

Q23. Does Dijkstra’s Algorithm work with negative edges?

No, it does not work with negative edge weights.

Q24. Which approach does Dijkstra’s Algorithm use?

Greedy approach.

Q25. What is relaxation in Dijkstra’s Algorithm?

Relaxation updates the shortest distance if a shorter path is found.

$d(u)+c(u,v)<d(v)$

Then:

$d(v)=d(u)+c(u,v)$

Q26. What is the initial distance of the source node in Dijkstra’s Algorithm?

The initial distance of the source node is 0.

Q27. What is the initial distance of other nodes in Dijkstra’s Algorithm?

Infinity (INF).

Bellman-Ford Algorithm

Q28. What is Bellman-Ford Algorithm?

Bellman-Ford Algorithm finds shortest paths in graphs with negative edge weights.

Q29. What is the main operation in Bellman-Ford Algorithm?

Relaxation of edges.

Q30. How many times are edges relaxed in Bellman-Ford Algorithm?

$V-1$ times, where V is the number of vertices.

Q31. How can a negative weight cycle be detected?

If any distance changes in the V-th iteration, then a negative weight cycle exists.

Q32. Does Bellman-Ford work with negative edges?

Yes.

Topological Sort

Q33. What is Topological Sorting?

It is a linear ordering of vertices in a DAG where u appears before v if there is an edge $u \to v$.

Q34. On which graph is topological sorting possible?

Directed Acyclic Graph (DAG).

Q35. Why is topological sorting impossible in cyclic graphs?

Because dependency ordering cannot be satisfied in cycles.

Q36. Give applications of topological sorting.

- Task scheduling
- Course prerequisite ordering
- Build systems

Kahn’s Algorithm

Q37. What is Kahn’s Algorithm?

Kahn’s Algorithm is a BFS-based algorithm for topological sorting.

Q38. Which vertices are inserted into the queue first in Kahn’s Algorithm?

Vertices with indegree 0.

Q39. What is indegree?

Indegree is the number of incoming edges of a vertex.

DFS Based Topological Sort

Q40. What data structure is used in DFS-based topological sorting?

Stack.

Q41. When is a node pushed into the stack in DFS topological sort?

After visiting all its adjacent vertices.

Backtracking

Q42. What is Backtracking?

Backtracking is a technique where solutions are built incrementally and invalid solutions are abandoned.

Q43. What happens when backtracking reaches a dead end?

The algorithm returns to the previous state and tries another path.

Q44. Name some applications of backtracking.

- Sudoku
- N Queens
- Graph Coloring
- Puzzle solving

Q45. What are the main steps of backtracking?

- Choose
- Explore
- Check validity
- Backtrack
- Repeat

Graph Coloring Problem

Q46. What is graph coloring?

Assigning colors to graph vertices so that adjacent vertices do not share the same color.

Q47. What is m-coloring?

Coloring a graph using at most m colors.

Q48. What is Chromatic Number?

The minimum number of colors needed to color a graph.

Q49. Give applications of graph coloring.

- Map coloring
- Timetable scheduling

N Queens Problem

Q50. What is the N Queens Problem?

It is the problem of placing N queens on an N×N chessboard so that no two queens attack each other.

Q51. Which positions must queens avoid?

- Same row
- Same column

- Same diagonal

Q52. Which technique is used to solve the N Queens Problem?

Backtracking.

Branch and Bound

Q53. What is Branch and Bound?

It is an optimization technique used to find optimal solutions by branching and pruning.

Q54. What is branching?

Dividing a problem into smaller subproblems.

Q55. What is pruning?

Eliminating branches that cannot produce better solutions.

Q56. What is bounding?

Computing upper or lower limits for subproblems.

Q57. Which search strategy is used in FIFO Branch and Bound?

Breadth First Search (BFS).

Approximation Algorithm

Q58. What is an approximation algorithm?

An algorithm that finds near-optimal solutions for NP-hard problems.

Q59. Why are approximation algorithms needed?

Because exact polynomial-time solutions are not available for NP-hard problems.

Q60. What is an approximate solution?

A feasible solution close to the optimal solution.

Traveling Salesman Problem (TSP)

Q61. What is the Traveling Salesman Problem?

The problem of visiting all cities exactly once and returning to the starting city with minimum cost.

Q62. What type of problem is TSP?

NP-hard problem.

Q63. What is Triangle Inequality?

$$d(A,C)\le d(A,B)+d(B,C)$$

The direct path should not cost more than an indirect path.

Q64. Which tree is used in TSP approximation?

Minimum Spanning Tree (MST).

Vertex Cover Problem

Q65. What is a vertex cover?

A subset of vertices such that every edge has at least one endpoint in the subset.

Q66. What is the goal of the Vertex Cover Problem?

To find the minimum size vertex cover.

Q67. Which graph is used in Vertex Cover Problem?

Undirected graph.

Q68. Give a real-life application of Vertex Cover.

Traffic camera placement.